Math. Z. 225, 611–624 (1997)

Mathematische Zeitschrift
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Boundedness properties of some convolution operators with singular measures*

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Received 21 April 1995; in final form 29 September 1995

1 Introduction

Let $Q = [-1,1]^{n-1}$ and let φ: $Q \to R$ be a measurable function ; suppose μ is the measure on R^n given by $\mu(E) = \int_Q \chi_E(x, \varphi(x))dx$, where dx denotes the Lebesgue measure on R^{n-1} and let T_μ be the convolution operator defined by $T_\mu f(x) = (\mu * f)(x)$ and let

$$E_\mu = \left\{ \left(\frac{1}{p}, \frac{1}{q}\right) : \|T_\mu\|_{L^p, L^q} < \infty, 1 \le p, q \le \infty \right\}$$

where the L^p spaces are taken with respect to the Lebesgue measure on R^n. The set E_μ is known in several cases. If the graph of φ has non zero Gaussian curvature at each point, a theorem of Littman (see [L]) implies that E_μ is the closed triangle with vertices (0, 0), (1,1) and $\left(\frac{n}{n+1}, \frac{1}{n+1}\right)$ (see [O]). Now, if the curvature vanishes in some point, E_μ can be strictly contained in the above triangle. Related examples in a more general context can be found in [O], and [R-S].

Our aim is to characterize the set E_μ in the case $n = 3$ and $\varphi(x_1, x_2) = |x_1|^\alpha + |x_2|^\beta$, $2 \le \alpha \le \beta$. We assume these hypothesis for the whole paper. Also, along this work, c will denote a positive constant not necessarily the same in a same chain of inequalities.

In Sect. 2 we find a convex polygonal region Σ such that $E_\mu \subset \Sigma$, we also obtain some estimates for the Fourier transform $\widehat{\mu}$. In Sect. 3 we get a complete description of E_μ in the cases $\alpha = 2$, $2 \le \beta \le 4$ and $2 \le \alpha = \beta \le 4$. Also we characterize the interior of E_μ and we obtain some additional information about its boundary in the other cases .

* Partially supported by CNR (Italy), Politecnico di Torino, CONICOR (Argentine) and SECITECOR (Argentine).

2 Auxiliary results

Let $Q = [-1,1]\times[-1,1]$ and let $\varphi : Q \to R$ be given by $\varphi(x_1,x_2) = |x_1|^{\alpha}+|x_2|^{\beta}$, with $\beta \geq \alpha \geq 2$. Suppose μ, E_μ are as in the introduction. The Riesz Thorin theorem implies that E_μ is a convex subset of the square $[0,1]\times[0,1]$. It is well known that if $\left(\frac{1}{p},\frac{1}{q}\right) \in E_\mu$ then p$\leq q$. (See [S-W] p. 33). We begin with the following lemmas.

Lemma 2.1. *If* $\left(\frac{1}{p},\frac{1}{q}\right) \in E_\mu$ *then* $\frac{1}{q} \geq \frac{3}{p} - 2$ *and* $\frac{1}{q} \geq \frac{1}{3p}$.

Proof. In order to prove the lemma, we take k such that $k \geq 4(1+2\alpha+2\beta)$ and we consider the function f=χ_{Q_δ} where $Q_\delta = [-\delta,\delta]^3$ and the set A_δ defined by

$$A_\delta = \{(x_1,x_2,x_3) : |x_1| \leq 1, |x_2| \leq 1, |\, x_3 - (|x_1|^{\alpha} + |x_2|^{\beta})\,| \leq \frac{\delta}{k}\}$$

We observe that, for some positive constant c (independent of δ) and for $x = (x_1,x_2,x_3) \in A_\delta$ we have $|(\mu * f)(x)| \geq c\delta^2$. Indeed

$$(\mu * f)(x) = \int_Q \chi_{Q_\delta}(x_1-y_1, x_2-y_2, x_3-\varphi(y_1,y_2))\, dy_1 dy_2$$

To estimate $|(\mu * f)(x)|$ we must measure the pairs $(y_1,y_2) \in Q$ satisfying

$$\left(y_1, y_2, |y_1|^{\alpha} + |y_2|^{\beta}\right) \in x + Q_\delta$$

For $|y_1 - x_1| \leq \frac{\delta}{k}$, $|y_2 - x_1| \leq \frac{\delta}{k}$ and $x \in A_\delta$, the mean value theorem implies

$$\left|x_3 - (|y_1|^{\alpha} + |y_2|^{\beta})\right| \leq$$

$$\leq \left|x_3 - (|x_1|^{\alpha} + |x_2|^{\beta})\right| + \left||x_1|^{\alpha} - |y_1|^{\alpha}\right| + \left|\,|x_2|^{\beta} - |y_2|^{\beta}\right| \leq$$

$$\leq \frac{\delta}{k}(1+2\alpha+2\beta) \leq \delta$$

Then $(x_1,x_2) + B_{\frac{\delta}{k}} \subset \left\{(y_1,y_2) : \left(y_1, y_2, |y_1|^{\alpha} + |y_2|^{\beta}\right) \in x + Q_\delta\right\}$, where $B_{\frac{\delta}{k}} = \left[-\frac{\delta}{k},\frac{\delta}{k}\right] \times \left[-\frac{\delta}{k},\frac{\delta}{k}\right]$. So, $|(\mu * f)(x)| \geq \frac{1}{4}\left(\frac{\delta}{k}\right)^2$ for $x \in A_\delta$. Now, $\|\mu * f\|_q \geq \left(\int_{A_\delta} |\mu * f|^q\right)^{\frac{1}{q}} \geq c\delta^{2+\frac{1}{q}}$. On the other hand, $\left(\frac{1}{p},\frac{1}{q}\right) \in E_\mu$ implies $\|\mu * f\|_q \leq$ $\leq c\,\|f\|_p = c\delta^{\frac{3}{p}}$ and so $\delta^{2+\frac{1}{q}} \leq c\delta^{-\frac{3}{p}}$ for all $\delta > 0$, then $\frac{1}{q} \geq \frac{3}{p} - 2$. Also it is easy to check that $\left(T_\mu\right)^* = T_{\mu^*}$ where

$$\mu^*(E) = \mu(-E) = \int_Q \chi_E(x_1,x_2,-\varphi(x_1,x_2))\, dx_1 dx_2.$$

Then $(\frac{1}{p},\frac{1}{q}) \in E_\mu$ if and only if $(\frac{1}{q'},\frac{1}{p'}) \in E_{\mu^*}$. The above argument applied to -φ tells us that $\frac{1}{q'} \geq \frac{3}{p'} - 2$ and so $\frac{1}{q} \geq \frac{1}{3p}$ $\square$

Lemma 2.2. *If* $\left(\frac{1}{p},\frac{1}{q}\right)\in E_\mu$ *then* $\frac{1}{q}\geq\frac{1}{p}-\frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}$.

Proof. Let f=χ_{Q_δ} where $Q_\delta=\left(-2\delta^\beta,2\delta^\beta\right)\times(-2\delta^\alpha,2\delta^\alpha)\times\left(-k\delta^{\alpha\beta},k\delta^{\alpha\beta}\right)$ with $k=2\alpha+2\beta+1$. We define

$$A_\delta=\{(x_1,x_2,x_3):|x_1|\leq\delta^\beta,|x_2|\leq\delta^\alpha,|\,x_3-(|x_1|^\alpha+|x_2|^\beta)\,|\leq\delta^{\alpha\beta}\}$$

We first show that $|(\mu*f)(x)|\geq c\delta^{\alpha+\beta}$ for some positive constant c (independent of δ) and all $x=(x_1,x_2,x_3)\in A_\delta$. Indeed,as before, we must measure the set of the pairs $(y_1,y_2)\in Q$ such that

$$\left(y_1,y_2,|y_1|^\alpha+|y_2|^\beta\right)\in x+Q_\delta$$

For $(y_1,y_2)\in\left(-\delta^\beta,\delta^\beta\right)\times(-\delta^\alpha,\delta^\alpha)$ we have $|y_1-x_1|\leq2\delta^\beta$and $|y_2-x_2|\leq 2\delta^\alpha$, also we have

$$\left||y_1|^\alpha+|y_2|^\beta-x_3\right|\leq$$

$$\leq|\,|y_1|^\alpha-|x_1|^\alpha\,|+|\,|y_2|^\beta-|x_2|^\beta\,|+|\,(|x_1|^\alpha+|x_2|^\beta)-x_3\,|$$

The mean value theorem gives us$|\,|x_1|^\alpha-|y_1|^\alpha\,|\leq2\alpha\delta^{\alpha\beta}$ and $|\,|x_2|^\beta-|y_2|^\beta\,|\leq 2\beta\delta^{\alpha\beta}$. So

$$\left||y_1|^\alpha+|y_2|^\beta-x_3\right|\leq(2\alpha+2\beta+1.)\,\delta^{\alpha\beta}.$$

Then $|(\mu*f)(x)|\geq c\delta^{\alpha+\beta}$ for $x\in A_\delta$. Now,

$$\|\mu*f\|_q\geq\left(\int_{A_\delta}|\mu*f|^q\right)^{\frac{1}{q}}\geq c\delta^{\alpha+\beta}\,|A_\delta|^{\frac{1}{q}}=c\delta^{\alpha+\beta+\frac{1}{q}(\alpha+\beta+\alpha\beta)}$$

But $\left(\frac{1}{p},\frac{1}{q}\right)\in E_\mu$ implies $\|\mu*f\|_q\leq c\,\|f\|_p=c\delta^{\frac{\alpha+\beta+\alpha\beta}{p}}$

So, $\delta^{(\alpha+\beta)(1+\frac{1}{q})+\frac{1}{q}\alpha\beta}\leq c\delta^{\frac{\alpha+\beta+\alpha\beta}{p}}$ for all $\delta>0$ small enough, then

$$\frac{1}{q}\geq\frac{1}{p}-\frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}.$$

and the lemma follows. □

Lemma 2.3. *If* $\left(\frac{1}{p},\frac{1}{q}\right)\in E_\mu$ *then* $\frac{1}{q}\geq\frac{2\beta+1}{\beta+1}\frac{1}{p}-1$.

Proof. Let $f=\chi_{Q_\delta}$ where $Q_\delta=\left(-\delta^\beta,\delta^\beta\right)\times(-2\delta,2\delta)\times\left(-k\delta^\beta,k\delta^\beta\right)$ with $k=2^{\alpha-1}\alpha+2\beta+1$. We define

$$A_\delta=\{(x_1,x_2,x_3):|x_1|\leq1,|x_2|\leq\delta,|\,x_3-(|x_1|^\alpha+|x_2|^\beta)\,|\leq\delta^\beta\}$$

We will show that there exists $c>0$ (independent of δ) such that $|(\mu*f)(x)|\geq c\delta^{\beta+1}$ for all $x\in A_\delta$ and for $\delta>0$ small enough . Indeed,as before, we must measure the pairs $(y_1,y_2)\in Q$ such that

$$\left(y_1, y_2, |y_1|^\alpha + |y_2|^\beta\right) \in x + Q_\delta$$

Now for $(y_1, y_2) \in \left(x_1 - \delta^\beta, x_1 + \delta^\beta\right) \times (-\delta, \delta)$ we have $|y_1 - x_1| \leq \delta^\beta$ and $|y_2 - x_2| \leq 2\delta$ also we have

$$\left||y_1|^\alpha + |y_2|^\beta - x_3\right| \leq$$

$$\leq| \ |y_1|^\alpha - |x_1|^\alpha \ | + | \ |y_2|^\beta - |x_2|^\beta \ | + | \ (|x_1|^\alpha + |x_2|^\beta) - x_3 \ |$$

As before taking account that $\delta < 1$, we have, for some t between $|x_1|$ and $|y_1|$

$$| \ |x_1|^\alpha - |y_1|^\alpha \ | \leq | \ x_1 - y_1 \ | \ \alpha t^{\alpha-1} \leq 2^{\alpha-1}\alpha\delta^\beta$$

Analogously we obtain, for some t between $|x_2|$ and $|y_2|$ $| \ |x_2|^\beta - |y_2|^\beta \ | \leq | \ x_2 - y_2 \ | \ \beta t^{\beta-1} \leq 2\beta\delta^\beta$. So $\left||y_1|^\alpha + |y_2|^\beta - x_3\right| \leq k\delta^\beta$.

Then $|(\mu * f)(x)| \geq c\delta^{\beta+1}$ for x$\in A_\delta$. Now,

$$\|\mu * f\|_q \geq \left(\int_{A_\delta} |\mu * f|^q\right)^{\frac{1}{q}} \geq c\delta^{\beta+1} \, |A_\delta|^{\frac{1}{q}} = c\delta^{\beta+1+\frac{1}{q}(1+\beta)}$$

But $\left(\frac{1}{p}, \frac{1}{q}\right) \in E_\mu$ implies $\|\mu * f\|_q \leq c \, \|f\|_p = c\delta^{\frac{2\beta+1}{p}}$

So, $\delta^{\beta+1+\frac{1}{q}(\beta+1)} \leq c\delta^{\frac{2\beta+1}{p}}$ for all positive and small enough δ. Then $\frac{1}{q} \geq \frac{2\beta+1}{\beta+1}\frac{1}{p} - 1$. □

Lemmas 2.1, 2.2, and 2.3 say that E_μ is contained in the convex polygonal region $\Sigma^{\alpha,\beta} \subset Q$ determined by the lines $L_0^{\alpha,\beta}$, L_1^β, L_2 with equations $\frac{1}{q} = \frac{1}{p} - \frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}$, $\frac{1}{q} = \frac{2\beta+1}{\beta+1}\frac{1}{p} - 1$ and $\frac{1}{q} = \frac{3}{p} - 2$ respectively, their symmetric with respect to the non principal diagonal and the principal diagonal $\frac{1}{q} = \frac{1}{p}$. A brief computation shows that Σ is either a trapezoid, or a pentagon or an hexagon according to $\alpha = \beta > \dot{2}$; $\alpha = 2$, $\beta > 2$ and $2 < \alpha < \beta$ respectively. From now away, let B_β, $C_{\alpha,\beta}$ be the points of intersection of L_1^β with L_2, and $L_0^{\alpha,\beta}$ with L_1^β respectively and let A_β be the intersection of L_1^β with the non principal diagonal $\frac{1}{q} + \frac{1}{p} = 1$, so $A_\beta = \left(\frac{2\beta+2}{3\beta+2}, \frac{\beta}{3\beta+2}\right)$, $B_\beta = \left(\frac{\beta+1}{\beta+2}, \frac{\beta-1}{\beta+2}\right)$ and $C_{\alpha,\beta} = \left(\frac{\alpha+\alpha\beta}{\alpha+\beta+\alpha\beta}, \frac{-\beta+\alpha\beta}{\alpha+\beta+\alpha\beta}\right)$. Also we put B'_β and $C'_{\alpha,\beta}$ for the symmetrics of B_β, $C_{\alpha,\beta}$ with respect to the non principal diagonal respectively, so $B'_\beta = \left(\frac{3}{\beta+2}, \frac{1}{\beta+2}\right)$ and $C'_{\alpha,\beta} = \left(\frac{\alpha+2\beta}{\alpha+\beta+\alpha\beta}, \frac{\beta}{\alpha+\beta+\alpha\beta}\right)$.

We prove now some estimates for the Fourier transform of μ given by $\widehat{\mu}(y) = \int e^{-i\langle x,y\rangle} d\mu(x)$. We will need the following

Remark 2.4. Proceeding as in the proof of Lemma 2.2 in [R-S], we obtain the following result.

For $k \in R$, $k > 1$, $\frac{1}{2} < Re\sigma < k$, a, b and $\lambda \in R$ we have

$$\left|\int_a^b e^{i\left(x^k - k\lambda x\right)} x^{\sigma-1} dx\right| \leq C_{k,\sigma} \left(1 + |\lambda|\right)^{-\left(\frac{k}{2} - Re\sigma\right)/(k-1)}$$

where, for k and Reσ fixed, $C_{k,\sigma}$ is polynomially increasing with Imσ.

Lemma 2.5. *The Fourier Transform $\widehat{\mu}$ satisfies*

$$|\widehat{\mu}(y_1, y_2, y_3)| \leq c\,|y_1|^{-\frac{\alpha-2}{2(\alpha-1)}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\left(\frac{1}{2(\alpha-1)}+\frac{1}{2(\beta-1)}\right)}$$

and

$$|\widehat{\mu}(y_1, y_2, y_3)| \leq c\,|y_3|^{-\frac{1}{\alpha}-\frac{1}{\beta}}$$

Proof. $\widehat{\mu}\ (y_1, y_2, y_3) = \int_Q e^{-i\langle (y_1,y_2,y_3),(x_1,x_2,|x_1|^\alpha+|x_2|^\beta)\rangle}dx_1dx_2 =$

$$= \int_{-1}^{1} e^{-i\left(y_1x_1+y_3|x_1|^\alpha\right)}dx_1 \cdot \int_{-1}^{1} e^{-i\left(y_2x_2+y_3|x_2|^\beta\right)}dx_2 = I_1I_2$$

To study I_1, we only estimate $\int_0^1$. Also, we suppose that y_3 is positive. The other cases reduce to this one. We can write

$$\left|\int_0^1 e^{-i\left(y_1x_1+y_3x_1^\alpha\right)}dx_1\right| = \left|y_3^{-\frac{1}{\alpha}}\int_0^{y_3^{\frac{1}{\alpha}}} e^{-i\left(y_1y_3^{-\frac{1}{\alpha}}x_1+x_1^\alpha\right)}dx_1\right|$$

Now, Remark 2.4 gives us

$$\left|y_3^{-\frac{1}{\alpha}}\int_0^{y_3^{\frac{1}{\alpha}}} e^{-i\left(y_1y_3^{-\frac{1}{\alpha}}x_1+x_1^\alpha\right)}dx_1\right| \leq cy_3^{-\frac{1}{\alpha}}\left(\frac{\alpha y_3^{\frac{1}{\alpha}}+|y_1|}{\alpha y_3^{\frac{1}{\alpha}}}\right)^{-\frac{\alpha-2}{2(\alpha-1)}} =$$

$$= cy_3^{-\frac{1}{\alpha}+\frac{\alpha-2}{2\alpha(\alpha-1)}}\cdot\left(y_3^{\frac{1}{\alpha}}+\alpha^{-1}|y_1|\right)^{-\frac{\alpha-2}{2(\alpha-1)}} \leq c.\min\left(y_3^{-\frac{1}{2(\alpha-1)}}\cdot|y_1|^{-\frac{\alpha-2}{2(\alpha-1)}}, y_3^{-\frac{1}{\alpha}}\right)$$

We can estimate I_2 in a similar way and so the lemma follows. □

Lemma 2.6. *If $\alpha = 2$, then*

$$|\widehat{\mu}(y_1, y_2, y_3)| \leq c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{\beta-1}}$$

Proof. We assume $y_3 > 0$. The other case follows from this one.

$$\widehat{\mu}\ (y_1, y_2, y_3) = \int_Q e^{-i\langle (y_1,y_2,y_3),(x_1,x_2,x_1^2+|x_2|^\beta)\rangle}dx_1dx_2 =$$

$$= \int_{-1}^{1} e^{-i\left(y_1x_1+y_3x_1^2\right)}dx_1\int_{-1}^{1} e^{-i\left(y_2x_2+y_3|x_2|^\beta\right)}dx_2 = I_1I_2.$$

We now estimate I_1.Van der Corput Lemma (See [S], p. 332) implies

$$|I_1| \leq \frac{c}{|y_3|^{\frac{1}{2}}} \tag{2.7}$$

Now $I_2 = \int_{-y_3^{\frac{1}{\beta}}}^{y_3^{\frac{1}{\beta}}} e^{-i\left(x_2 y_2 y_3^{-\frac{1}{\beta}} + |x_2|^\beta\right)} y_3^{-\frac{1}{\beta}} dx_2$. Remark 2.4 implies

$$|I_2| \leq c\,|y_3|^{-\frac{1}{2(\beta-1)}} \left(|y_3|^{\frac{1}{\beta}} + |y_2|\right)^{-\frac{\beta-2}{2(\beta-1)}} \leq c\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{2(\beta-1)}} \tag{2.8}$$

Suppose $\left|\frac{y_1}{2y_3}\right| \leq 2$, from (2.7) and (2.8) we obtain

$$|\widehat{\mu}\,(y_1, y_2, y_3)| \leq c\,|y_3|^{-\frac{1}{2}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{2(\beta-1)}} \leq$$

$$\leq c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{\beta-1}}$$

In the case $\left|\frac{y_1}{2y_3}\right| > 2$, we set $\psi(x) = xy_1 + x^2 y_3$ then $\psi'(x) = y_1 + 2xy_3 = 2y_3\left(\frac{y_1}{2y_3} + x\right)$, so for x$\in [-1, 1]$ we have $|\psi'(x)| \geq \frac{|y_1|}{2}$. Van der Corput Lemma gives us

$$|I_1| \leq \frac{c}{|y_1|}$$

Also $|I_1| \leq c$, so we have

$$|I_1| \leq c\,|y_1|^{-\frac{1}{2}} = c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}-\frac{1}{2(\beta-1)}} \leq c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{2(\beta-1)}}$$

and so (2.8) gives us

$$|\widehat{\mu}\,(y_1, y_2, y_3)| \leq c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{\beta-1}}$$

and the proof is over. □

Lemma 2.9. *Suppose $0 < a$, $0 < d < h$, suppose either $R = [-a, a] \times [d, h]$ or $R = [-a, a] \times [-h, -d]$ and let $\rho^{\pm}$ be the measure defined by*

$$\rho^{\pm}(E) = \int_R \chi_E(x_1, x_2, \pm\varphi(x_1, x_2))dx_1 dx_2$$

then there exists a positive constant c such that

$$\left|\widehat{\rho^{\pm}}\,(y_1, y_2, y_3)\right| \leq c\,|y_1|^{-\frac{\alpha-2}{2(\alpha-1)}}\,|y_2|^{-\frac{\alpha-2}{2(\alpha-1)}}\,|y_3|^{-\frac{1}{\alpha-1}}$$

and

$$\left|\widehat{\rho^{\pm}}\,(y_1, y_2, y_3)\right| \leq c\,|y_3|^{-\frac{1}{2}-\frac{1}{\alpha}}$$

Proof. We prove the estimate for $R = [-a, a] \times [d, h]$ and ρ^{+}. The other cases are analogous

$$\widehat{\rho^{+}}\ (y_1, y_2, y_3) = \int_R e^{-i\langle (y_1, y_2, y_3), (x_1, x_2, |x_1|^\alpha + |x_2|^\beta)\rangle} dx_1 dx_2 =$$

$$= \int_{-a}^{a} e^{-i(y_1 x_1 + y_3 |x_1|^\alpha)} dx_1 \int_d^h e^{-i\left(y_2 x_2 + y_3 x_2^\beta\right)} dx_2 = I_1 I_2.$$

We now estimate I_2. Since $x_2 > d > 0$, Van der Corput Lemma (See [S], p. 332) implies that

$$|I_2| \leq \frac{c}{|y_3|^{\frac{1}{2}}}$$

For $\left|\frac{y_2}{\beta y_3}\right| > 2h^{\beta-1}$ and $x \in [d,h]$ we set $\psi(x) = xy_2 + x^{\beta}y_3$. So $\psi'(x) = y_2 + \beta x^{\beta-1}y_3 = = \beta y_3\left(\frac{y_2}{\beta y_3} + x^{\beta-1}\right)$ and then $|\psi'(x)| \geq \frac{|y_2|}{2}$, thus Van der Corput Lemma gives us $|I_2| \leq \frac{c}{|y_2|}$. Also $|I_2| \leq h - d$ then $|I_2| \leq c\,|y_2|^{-\frac{1}{2}}$. Now, as in (2.8) we find $|I_1| \leq c\,|y_3|^{-\frac{1}{2(\alpha-1)}}\,|y_1|^{-\frac{\alpha-2}{2(\alpha-1)}}$ and $|I_1| \leq |y_3|^{-\frac{1}{\alpha}}$. So we obtain for I_1 and I_2 the same kind of bounds than those obtained in the proof of the Lemma 2.6 and then , with the same proof the lemma follows. □

Remark 2.10. In a similar way as in Lemma 2.9 we can prove

Suppose $0 < a$, $0 < d < h$, suppose either $R = [d,h] \times [-a,a]$ or $R = [-h,-d] \times [-a,a]$, and let $\rho^{\pm}$ be the measure defined by

$$\rho^{\pm}(E) = \int_R \chi_E(x_1, x_2, \pm\varphi(x_1,x_2))dx_1dx_2$$

then there exists a positive constant c such that for $y = (y_1,y_2,y_3) \in R^3$

$$\left|\widehat{\rho^{\pm}}(y)\right| \leq c\,|y_1|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_2|^{-\frac{\beta-2}{2(\beta-1)}}\,|y_3|^{-\frac{1}{\beta-1}}$$

and

$$\left|\widehat{\rho^{\pm}}(y)\right| \leq |y_3|^{-\frac{1}{2}-\frac{1}{\beta}}$$

3 The main results

Lemma 3.1. *Suppose* $0 \leq \gamma < \frac{1}{2}$ *and* $\tau \geq 0$. *Let* $B \subset R^2$ *be a rectangle and* $\nu^{\pm}$ *the finite Borel measures on* R^3 *given by*

$$\nu^{\pm}(E) = \int_B \chi_E(x_1,x_2,\pm\varphi(x_1,x_2))\,dx_1dx_2$$

suppose

$$\left|\widehat{\nu^{\pm}}(y_1,y_2,y_3)\right| \leq c\,|y_1|^{-\gamma}\,|y_2|^{-\gamma}\,|y_3|^{-\gamma-\tau}$$

then the convolution operators defined by $Tf = \nu^{\pm} * f$ *are bounded from* $L^{\frac{2(1+\tau)}{2\tau+2\gamma+1}}\left(R^3\right)$ *into* $L^{2(1+\tau)}\left(R^3\right)$.

Proof. We study ν^{+}, the other case is analogous. For $Re(z) > 0$ and $t \in R$ we consider the fractional integration kernel

$$I_z(t) = \frac{2^{-\frac{z}{2}}}{\Gamma\left(\frac{z}{2}\right)}\,|t|^{z-1}$$

and its analytic extension to $z \in C$. In particular we have $\widehat{I_z} = I_{1-z}$, also $I_0 = c\delta$ where δ denotes the Dirac distribution at the origin. We also define J_z and H_z as the distributions on R^3 given by the tensor products $J_z = \delta \otimes \delta \otimes I_z$ and H_z=1⊗1 ⊗ I_z. So for $Re\,(z) > 0$,

$$H_z\,(x_1, x_2, x_3) = \frac{2^{-\frac{z}{2}}}{\Gamma\left(\frac{z}{2}\right)}\,|x_3|^{z-1}$$

Moreover we have that $\widehat{H_z} = J_{1-z}$.

Let $T_z f = f * \nu^+ * J_z$ for $z \in C$. We observe that $T_z f = \left(\widehat{f\nu^+} H_{1-z}\right)^\vee$. We first assume $\tau > 0$. We consider the strip $-\tau \leq Re\,(z) \leq 1$. For $Re\,(z) = 1$ we have

$$\|T_z f\|_\infty = \|f * \nu^+ * J_z\|_\infty \leq \|f\|_1 \,\|\nu^+ * J_z\|_\infty .$$

Now, $\nu^+ * J_z$ is given by the function

$$\left(\nu^+ * J_z\right)(x_1, x_2, x_3) = \frac{2^{-\frac{z}{2}}}{\Gamma\left(\frac{z}{2}\right)} \left|x_3 - \left(|x_1|^\alpha + |x_2|^\beta\right)\right|^{iIm(z)} \chi_B\,(x_1, x_2)$$

and so $\|\nu^+ * J_z\|_\infty \leq \frac{c}{\left|\Gamma\left(\frac{z}{2}\right)\right|}$. Then for $Re\,(z) = 1$, we find $\|T_z\|_{1,\infty} \leq c\left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}$.

For $Re\,(z) = -\tau$ we have $\|T_z\|_{\frac{2}{2\gamma+1},2} \leq \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|}$. Indeed, for $y = (y_1, y_2, y_3) \in R^3$ we have

$$\left|(T_z f)^\wedge(y)\right| = \left|\widehat{\nu^+}\,(\delta \otimes \delta \otimes I_z)^\wedge(y)\right| \left|\widehat{f}\,(y)\right| \leq$$

$$\leq \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|} \frac{|y_3|^\tau}{|y_1|^\gamma\,|y_2|^\gamma\,|y_3|^{\gamma+\tau}} \left|\widehat{f}\,(y)\right| = \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|} \frac{1}{|y_1|^\gamma\,|y_2|^\gamma\,|y_3|^\gamma} \left|\widehat{f}\,(y)\right|$$

Then $\|T_z f\|_2 = \left\|(T_z f)^\wedge\right\|_2 \leq \left\|(Kf)^\wedge\right\|_2$ where

$$Kf = \left(\left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1} |y_1|^{-\gamma}\,|y_2|^{-\gamma}\,|y_3|^{-\gamma} \left|\widehat{f}\,(y)\right|\right)^\vee$$

K can be seen as a tensor product H⊗H⊗H where H convolves $L^{\frac{2}{2\gamma+1}} \to L^2$, so

$$\|T_z f\|_2 \leq \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|} \|f\|_{\frac{2}{2\gamma+1}}$$

It is easy to see, with the aid of the Stirling formula, that the family $\{T_z\}$ satisfies the hypothesis of the complex interpolation theorem (see [S-W], p 205) and so $T_\nu = cT_0$ is bounded from $L^{\frac{2(1+\tau)}{2\tau+2\gamma+1}}\left(R^3\right)$ into $L^{2(1+\tau)}\left(R^3\right)$. We note that if $\tau = 0$ and $Re\,(z) = 0$ we can obtain, as above

$$\|T_z f\|_2 \leq \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|} \|f\|_{\frac{2}{2\gamma+1}}$$

in particular $\|T_{\nu^+}\|_{\frac{2}{2\gamma+1},2} < \infty$. □

Theorem 3.2. *For $\alpha = 2$ and $2 < \beta \leq 4$, E_μ is the closed pentagon with vertices $(0,0), (1,1), A_\beta, B_\beta,$ and B'_β.*

Proof. Lemmas 2.1, and 2.3 together with the convexity of E_μ and duality considerations imply that we must only check that A_β and B_β belong to E_μ. Lemmas 2.6 and 3.1 imply that $B \in E_\mu$. We consider, for $-\frac{\beta+2}{2\beta} \leq Re\,(z) \leq 1$, the analytic family of operators $\{T_z\}$ defined as in the proof of the Lemma 3.1. So, for $Re\,(z) = 1$, $\|T_z f\|_{1,\infty} \leq c \left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}$.

For $Re\,(z) = -\frac{\beta+2}{2\beta}$, we have $\|T_z\|_{2,2} \leq c \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1}$. Indeed

$$\|T_z f\|_2 = \left\|\widehat{f}\widehat{\mu}H_{1-z}\right\|_2 \leq \|\widehat{\mu}H_{1-z}\|_\infty \|f\|_2$$

Now Lemma 2.5 implies $|\widehat{\mu}(y_1, y_2, y_3)| \leq c\,|y_3|^{-\frac{\beta+2}{2\beta}}$, on the other hand

$$\left|H_{1-z}\left(y_1, y_2, y_3\right)\right| = \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|}\,|y_3|^{\frac{\beta+2}{2\beta}}$$

So $\|\widehat{\mu}H_{1-z}\|_\infty \leq c \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1}$. Then we can apply the complex interpolation theorem to obtain that $T_\mu = cT_0$ is bounded from $L^{\frac{3\beta+2}{2\beta+2}}$ into $L^{\frac{3\beta+2}{\beta}}$. □

Theorem 3.3. *For $2 < \alpha = \beta \leq 4$, E_μ is the closed trapezoid with vertices (0,0), (1,1), B_β and B'_β.*

Proof. Lemmas 2.1 and 2.2 together with the convexity of E_μ and duality considerations imply that we must only check that B_β belongs to E_μ and this follows from Lemmas 2.5 and 3.1. □

Proposition 3.4. *Suppose $0 < a$, $0 < d < h$, $\alpha \leq 4$. Suppose either $R = [-a, a] \times [d, h]$ or $R = [-a, a] \times [-h, -d]$, and $\rho^{\pm}$ be defined as in Lemma 2.9. Then $E_{\rho^{\pm}}$ contains the closed polygon with vertices (0,0), (1,1), A_α, B_α B'_α.*

Proof. We prove the statement for ρ^+ and $R = [-a, a] \times [d, h]$. The other cases are analogous. In order to check that $A_\alpha \in E_{\rho^+}$, we consider the analytic family of operators $\{T_z\}$, defined by $T_z f = f * \rho^+ * J_z$ for $Re\,(z) > 0$ and $T_z f = \left(\widehat{f}\widehat{\rho^+}H_{1-z}\right)^\vee$ for $Re\,(z) \leq 0$, with H_z and J_z defined as in the proof of the Lemma 3.1. Now we take the strip $-\frac{\alpha+2}{2\alpha} \leq Re\,(z) \leq 1$. For $Re\,(z) = 1$, we find, as in Lemma 3.1, $\|T_z\|_{1,\infty} \leq c \left|\Gamma\left(\frac{z}{2}\right)\right|^{-1}$.

For $Re\,(z) = -\frac{\alpha+2}{2\alpha}$, $\|T_z f\|_{2,2} \leq c \left|\Gamma\left(\frac{1-z}{2}\right)\right|^{-1}$. Indeed

$$\|T_z f\|_2 = \left\|\widehat{f}\widehat{\rho^+}H_{1-z}\right\|_2 \leq \left\|\widehat{\rho^+}H_{1-z}\right\|_\infty \|f\|_2$$

Now Lemma 2.9 implies that $\left|\widehat{\rho^+}(y_1, y_2, y_3)\right| \leq c\,|y_3|^{-\frac{\alpha+2}{2\alpha}}$. On the other hand

$$\left|H_{1-z}\left(y_1, y_2, y_3\right)\right| = \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|}\,|y_3|^{\frac{\alpha+2}{2\alpha}}$$

So $\left\|\widehat{\rho^+}H_{1-z}\right\|_\infty \leq \frac{c}{\left|\Gamma\left(\frac{1-z}{2}\right)\right|}$. Then we can apply the complex interpolation theorem to obtain that T_{ρ^+} is bounded from $L^{\frac{3\alpha+2}{2\alpha+2}}$ into $L^{\frac{3\alpha+2}{\alpha}}$. On the other hand Lemmas 2.9 and 3.1 imply that $B_\alpha \in E_{\rho^+}$. Now, $\left(T_{\rho^+}\right)^* = T_{(\rho^+)^*}$ where $(\rho^+)^*(E) = \int_{-R} \chi_E(x_1, x_2, -\varphi(x_1, x_2))\, dx_1 dx_2$. The above argument implies that $\left(\frac{\alpha+1}{\alpha+2}, \frac{\alpha-1}{\alpha+2}\right) \in E_{(\rho^+)^*}$, and so $\left(\frac{3}{\alpha+2}, \frac{1}{\alpha+2}\right) \in E_{\rho^+}$. □

In a similar way we get the following

Remark 3.5. Suppose $0 < a$, $0 < d < h$, $\beta \leq 4$ and $\rho^\pm$ defined as in Lemma 2.9. Suppose either $R = [d, h] \times [-a, a]$ or $R = [-h, -d] \times [-a, a]$. Then $E_{\rho^\pm}$ contains the closed polygon with vertices $(0,0)$, $(1,1)$, A_β, B_β and B'_β.

Theorem 3.6. *For $2 < \alpha < \beta \leq 4$, E_μ is the closed hexagon with vertices $(0,0)$, $(1,1)$, B_β, $C_{\alpha,\beta}$, $C'_{\alpha,\beta}$ and B'_β, except perhaps the closed segment $C_{\alpha,\beta}C'_{\alpha,\beta}$.*

Proof. For $k \in N \cup \{0\}$, let

$$U_k = \left\{(x_1, x_2) \in R^2 : 2^{-k\alpha\beta} < |x_1|^\alpha + |x_2|^\beta \leq 2^{(-k+1)\alpha\beta}\right\}$$

and let

$$U = \left\{(x_1, x_2) \in R^2 : |x_1|^\alpha + |x_2|^\beta \leq 2^{\alpha\beta}\right\}.$$

We observe that, since $Q \subset U$, then $\mu \leq \nu$, where ν is the measure given by $\nu(E) = \int_U \chi_E(x_1, x_2, \varphi(x_1, x_2)) dx_1 dx_2$, so, recalling Lemmas 2.1, 2.2 and 2.3, it is enough to prove the assertion of the theorem for the operator $T_\nu f = \nu * f$. Now $\nu = \sum_{k\geq 0} \nu_k$ where $\nu_k = \int_{U_k} \chi_E(x_1, x_2, \varphi(x_1, x_2)) dx_1 dx_2$. We put $V = \left\{(x_1, x_2) \in R^2 : |x_1|^\alpha + |x_2|^\beta = 1\right\}$. So $V = V_1^+ \cup V_1^- \cup V_2^+ \cup V_2^-$ where

$$V_1^\pm = \left\{(x_1, x_2) \in V : |x_1| \leq 2^{-\frac{1}{\alpha}}, x_2 \in R^\pm\right\}$$

$$V_2^\pm = \left\{(x_1, x_2) \in V : |x_2| \leq 2^{-\frac{1}{\beta}}, x_1 \in R^\pm\right\}$$

We consider on R^2 and R^3 the dilations defined by $\delta \cdot (x_1, x_2) = \left(\delta^\beta x_1, \delta^\alpha x_2\right)$ and $\delta \cdot (x_1, x_2, x_3) = \left(\delta^\beta x_1, \delta^\alpha x_2, \delta^{\alpha\beta} x_3\right)$. Then $U_k = U_{1,k}^+ \cup U_{1,k}^- \cup U_{2,k}^+ \cup U_{2,k}^-$ with $U_{i,k}^\pm = \left\{\delta \cdot (x_1, x_2) : (x_1, x_2) \in V_i^\pm, 2^{-k} < \delta \leq 2^{1-k}\right\}$, $i = 1, 2$. Also $U_{i,k}^\pm = 2^{-k} \cdot U_{i,0}^\pm$. We write $\nu_k = \nu_{1,k}^+ + \nu_{1,k}^- + \nu_{2,k}^+ + \nu_{2,k}^-$ where

$$\nu_{i,k}^\pm(E) = \int_{U_{i,k}^\pm} \chi_E(x_1, x_2, \varphi(x_1, x_2)) dx_1 dx_2$$

For $f : R^3 \to R$, $\delta > 0$ we set $(\delta \cdot f)(x) = f(\delta \cdot x)$. So, by a change of variable we obtain, for $i = 1, 2$

$$T_{\nu_{i,k}^\pm} f(x) = 2^{-k(\alpha+\beta)} T_{\nu_{i,0}^\pm}\left(2^{-k} \cdot f\right)\left(2^k \cdot x\right)$$

and then for $1 < p, q < \infty$

$$\left\|T_{\nu^{\pm}_{i,,k}} f\right\|_q = \left(\int_{R^3} \left|T_{\nu^{\pm}_{i,k}} f(x)\right|^q dx\right)^{\frac{1}{q}} =$$

$$= 2^{-k(\alpha+\beta)} \left(\int_{R^3} \left|T_{\nu^{\pm}_{i,,0}} \left(2^{-k} \cdot f\right)\left(2^k \cdot x\right)\right|^q dx\right)^{\frac{1}{q}} =$$

$$= 2^{-k\left((\alpha+\beta)+\frac{1}{q}(\alpha+\beta+\alpha\beta)\right)} \left(\int_{R^3} \left|T_{\nu^{\pm}_{i,0}} \left(2^{-k} \cdot f\right)(x)\right|^q dx\right)^{\frac{1}{q}} \leq$$

$$\leq 2^{-k\left((\alpha+\beta)+\frac{1}{q}(\alpha+\beta+\alpha\beta)\right)} \left\|T_{\nu^{\pm}_{i,0}}\right\|_{p,q} \left\|2^{-k} \cdot f\right\|_p =$$

$$2^{-k\left((\alpha+\beta)+\frac{1}{q}(\alpha+\beta+\alpha\beta)-\frac{1}{p}(\alpha+\beta+\alpha\beta)\right)} \left\|T_{\nu^{\pm}_{i,0}}\right\|_{p,q} \|f\|_p =$$

$$= 2^{-k(\alpha+\beta+\alpha\beta)\left(\frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}+\frac{1}{q}-\frac{1}{p}\right)} \left\|T_{\nu^{\pm}_{i,,0}}\right\|_{p,q} \|f\|_p$$

Since $(\alpha+\beta+\alpha\beta)\left(\frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}+\frac{1}{q}-\frac{1}{p}\right) > 0$ for $\frac{1}{q} > \frac{1}{p} - \frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}$, in order to obtain the theorem, it is enough to prove that $E_{\nu^{\pm}_{1,0}}$ and $E_{\nu^{\pm}_{2,0}}$ contain the closed pentagons with vertices $(0,0)$, $(1,1)$, A_α, B_α, B_α and $(0,0)$, $(1,1)$, A_β, B_β, B'_β respectively. We will check this assertion for $E_{\nu^{+}_{1,0}}$. The other cases are analogous. We observe that $U^{+}_{1,0}$ is contained in the rectangle $R = \left[-2^{\beta-\frac{1}{\alpha}}, 2^{\beta-\frac{1}{\alpha}}\right] \times \left[2^{-\frac{1}{\beta}}, 2^{\alpha}\right]$. Let ρ be the measure defined by

$$\rho(E) = \int_R \chi_E(x_1, x_2, \varphi(x_1, x_2)) dx_1 dx_2$$

then $\nu^{+}_{1,0} \leq \rho$. So, in order to obtain the stated result for $E_{\nu^{+}_{1,0}}$, it is enough to show that E_ρ contains the closed pentagon with vertices $(0,0)$, $(1,1)$, A_α, B_α, B'_α and this follows from Proposition 3.4. □

Theorems 3.2 and 3.3 give a complete description of E_μ in the cases $2{=}\alpha < \beta \leq 4$ and $2 < \alpha = \beta \leq 4$. From now on we obtain some theorems that give less information about E_μ but that still hold even if $\beta > 4$.

Theorem 3.7. *For* $2{=}\alpha < \beta$, E_μ *is the pentagon with vertices* $(0,0)$, $(1,1)$, A_β, B_β, *and* B'_β, *except perhaps* B_β, B'_β *and the open segments with vertices* A_β, B_β *and* B'_β, A_β *respectively*.

Proof. Lemma 2.5 gives us $|\widehat{\mu}(y_1, y_2, y_3)| \leq c\,|y_3|^{-\frac{\beta+2}{2\beta}}$. Now we consider the analytic family of operators $\{T_z\}$, $-\frac{\beta+2}{2\beta} \leq Re(z) \leq 1$, defined as in the proof of the Lemma 3.1 to obtain, as there, that $A_\beta \in E_\mu$.

We write $\mu = \mu^{(1)} + \mu^{(2)} + \mu^{(3)} + \mu^{(4)}$ where $\mu^{(i)}$ is the measure μ restricted to the $i-$th. quadrant of Q. We treat $\mu^{(1)}$, the other cases are similar. We put $Q_j = [0,1] \times \left[2^{-j}, 2^{-j+1}\right]$, $j \in N$. Let $\mu^{(1)}_j$ be the restriction of $\mu^{(1)}$ to Q_j, so $\mu^{(1)} = \sum_{j\geq 0} \mu^{(1)}_j$. Now

$$\widehat{\mu_j^{(1)}}\ (y_1,y_2,y_3)=\int_0^1 e^{-i\left(y_1x_1+y_3x_1^2\right)}dx_1\cdot\int_{2^{-j}}^{2^{-j+1}} e^{-i\left(y_2x_2+y_3x_2^\beta\right)}dx_2=I_1I_2$$

The Van Der Corput Lemma applied to I_1 and I_2 yields us $|I_1|\leq c\,|y_3|^{-\frac{1}{2}}$ and $|I_2|\leq c2^{j\frac{\beta-2}{2}}\,|y_3|^{-\frac{1}{2}}$, so

$$\left|\widehat{\mu_j^{(1)}}\,(y_1,y_2,y_3)\right|\leq c2^{j\frac{\beta-2}{2}}\,|y_3|^{-1} \tag{3.8}$$

We consider, for $-1\leq Re\,(z)\leq 1$ the analytic family of operators $U_z^jf=e^{z^2}.\mu_j*J_z*f$, where J_z is defined as in the proof of the Lemma 3.1. Now, for $Re\,(z)=-1$, $H_{1-z}\,(y_1,y_2,y_3)=\frac{2^{-\frac{1-z}{2}}}{\Gamma\left(\frac{1-z}{2}\right)}\,|y_3|^{-z}$ and so (3.8) implies

$$\left\|U^j_{-1+i\gamma}\right\|_{2,2}\leq ce^{(-1+i\gamma)^2}2^{j\frac{\beta-2}{2}}\left|\Gamma\left(1-\frac{i\gamma}{2}\right)\right|^{-1}\leq c2^{j\frac{\beta-2}{2}}$$

Also

$$\left\|U^j_{1+i\gamma}\right\|_{1,\infty}\leq ce^{(1+i\gamma)^2}\left|\Gamma\left(\frac{1+i\gamma}{2}\right)\right|^{-1}\leq c$$

The analytic family U_z^j satisfies the hypothesis of the complex interpolation theorem and so $\left\|T_{\mu_j^{(1)}}\right\|_{\frac{4}{3},4}\leq c2^{j\frac{\beta-2}{4}}$ (see [S-W], p. 209). Now, $\left\|T_{\mu_j^{(1)}}\right\|_{1,1}=\left\|\mu_j^{(1)}\right\|=2^{-j}$, and so, for $\left(\frac{1}{p},\frac{1}{q}\right)$ in the open segment between $(1,1)$ and B_β, the Riesz Thorin theorem gives us $\left\|T_{\mu_j^{(1)}}\right\|_{p,q}\leq 2^{-j\epsilon}$, for some $\epsilon>0$ and all j. So $\sum_{j\geq 0}\left\|T_{\mu_j^{(1)}}\right\|_{p,q}<\infty$. □

Theorem 3.9. *For $2<\alpha=\beta$, E_μ is the trapezoid with vertices $(0,0)$, $(1,1)$, B_β, and B'_β, except perhaps the closed segment $B_\beta B'_\beta$.*

Proof. Let Σ' the closed pentagon with vertices $(0,0)$, $(1,1)$, A_β, B_β, and B'_β, without the closed segments $A_\beta B_\beta$ and $B'_\beta A_\beta$. Let $\nu^{\pm}_{i,,0}$, ρ be defined as in the proof of the Theorem 3.6. We can see, as there, that for $\frac{1}{q}>\frac{1}{p}-\frac{2\beta}{2\beta+\beta^2}$, the assumption $\left\|T_{\nu^{\pm}_{i,,0}}\right\|_{p,q}<\infty$ for $i=1,2$ implies $\|T_\mu\|_{p,q}<\infty$. So, it is enough to show that if $\left(\frac{1}{p},\frac{1}{q}\right)$ belongs to Σ' then $\left\|T_{\nu^{\pm}_{i,,0}}\right\|_{p,q}<\infty$ for $i=1,2$.

We first study $T_{\nu^+_{1,0}}$. Lemma 2.9 gives us

$$|\widehat{\rho}(y_1,y_2,y_3)|\leq|y_3|^{-\frac{1}{2}-\frac{1}{\beta}}$$

So, as in the proof of the Theorem 3.7, we obtain $A_\beta\in E_\rho\subset E_{\nu^+_{1,0}}$. We set $F_j=\left\{t\in R:2^{-j\left(\beta-\frac{1}{\beta}\right)}\leq|t|\leq 2^{(-j+1)\left(\beta-\frac{1}{\beta}\right)}\right\}$ and $G_j=F_j\times\left[2^{-\frac{1}{\beta}},2^\beta\right]$. Let σ_j be the measure defined by

$$\sigma_j(E) = \int_{G_j} \chi_E\left(x_1, x_2, \varphi(x_1, x_2)\right) dx_1 dx_2$$

So we can write $\rho = \sum_{j\geq 0} \sigma_j$

Now

$$\widehat{\sigma_j}(y_1, y_2, y_3) =$$

$$\int_{F_j} e^{-i\left(y_1 x_1 + y_3 x_1^{\beta}\right)} dx_1 \int_{2^{-\beta}}^{2^{\beta}} e^{-i\left(y_2 x_2 + y_3 x_2^{\beta}\right)} dx_2 = I_1 I_2$$

The Van Der Corput Lemma applied to estimate I_1 and I_2 yields us $|I_2| \leq c\,|y_3|^{-\frac{1}{2}}$ and $|I_1| \leq c 2^{j\frac{\beta-2}{2}\left(\beta-\frac{1}{\beta}\right)} |y_3|^{-\frac{1}{2}}$, so

$$|\widehat{\sigma_j}(y_1, y_2, y_3)| \leq c 2^{j\frac{\beta-2}{2}\left(\beta-\frac{1}{\beta}\right)} |y_3|^{-1}$$

As in the proof of the Theorem 3.7 we obtain $\left\|T_{\sigma_j}\right\|_{\frac{4}{3},4} \leq c 2^{j\frac{\beta-2}{4}\left(\beta-\frac{1}{\beta}\right)}$. and $\left\|T_{\sigma_j}\right\|_{1,1} = \|\sigma_j\| = c 2^{-j\left(\beta-\frac{1}{\beta}\right)}$. So, as in 3.7, $T_\rho = \sum_{j\geq 0} T_{\sigma_j}$ is bounded from L^p into L^q if $\left(\frac{1}{p}, \frac{1}{q}\right)$ belongs to Σ', so the same holds for $T_{\nu^+_{1,0}}$. The treatment of $T_{\nu^-_{1,0}}$ is analogous. We proceed in a similar way with $T_{\nu^{\pm}_{2,0}}$ to obtain that $T_{\nu^{\pm}_{2,0}}$ is bounded from L^p into L^q if $\left(\frac{1}{p}, \frac{1}{q}\right)$ belongs to Σ'. □

Theorem 3.10. *For $2< \alpha < \beta$, E_μ is the closed hexagon with vertices (0,0), (1,1), B_β, $C_{\alpha,\beta}$, $C'_{\alpha,\beta}$ and B'_β except perhaps the closed segments $B_\beta C_{\alpha,\beta}$, $C_{\alpha,\beta}C'_{\alpha,\beta}$ and $B'_\beta C'_{\alpha,\beta}$.*

Proof. Let $\nu^{\pm}_{i,0}$, U be defined as in the proof of the Theorem 3.6. As there, if $\frac{1}{q} > \frac{1}{p} - \frac{\alpha+\beta}{\alpha+\beta+\alpha\beta}$, then $\left\|T_{\nu^{\pm}_{i,0}}\right\|_{p,q} < \infty$ for $i = 1, 2$ implies $\|T_\mu\|_{p,q} < \infty$. But it is easy to see, as in the proof of the Theorem 3.9 that $\left\|T_{\nu^{\pm}_{i,0}}\right\|_{p,q} < \infty$ for $i = 1, 2$ if $\left(\frac{1}{p}, \frac{1}{q}\right)$ belongs to the pentagon with vertices $(0,0)$, $(1,1)$, A_β, B_β, B'_β without the closed segments $A_\beta B_\beta$ and $A_\beta B'_\beta$. □

Remark 3.11. Let S be the graph of $\varphi : Q \to R$ where $\varphi(x_1, x_2) = |x_1|^{\alpha} + |x_2|^{\beta}$, with $2\leq\alpha, \beta$, and let σ the euclidean area measure on S, let Ψ a measurable function on S satisfying $c_1 \leq \Psi \leq c_2$,for some positive constants c_1 and c_2. We consider the measure ν on R^3 given by

$$\nu(E) = \int \chi_{E\cap S}(y)\, \Psi(y)\, d\sigma(y)$$

So, we have, for some c and $c' > 0$ and all positive and measurable function $f : S \to R$

$$cT_\mu f \leq T_\nu f \leq c' T_\mu f,$$

then all the results stated for T_μ hold for T_ν.

Acknowledgement. We are deeply indebted to Prof. Fulvio Ricci who suggested this problem and generously guided us.

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